THE MULTIPLICITY OF A TWO-PERIODIC COMPLEX OF FINITE LENGTH MODULES IS THE ALTERNATING SUM OF THE LENGTHS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed. The author leaves the two reduced cases to the reader; that sentence is part of the author's text and is reproduced as written.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter referenced lemma is located at its \label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Two-periodic complexes and their multiplicity e_R. Same chapter, Definition environment labelled definition-periodic-complex (source line 110) together with the displayed definition of e_R at source lines 200-205 (label equation-multiplicity-coker-ker, line 203). Supplies e_R(M, N, phi, psi) and the (2,1)-periodic specialisation e_R(M, phi, psi); the reduction to the two extreme cases is the counted content.
- Additivity of the multiplicity in short exact sequences. Same chapter, Lemma labelled lemma-additivity-periodic-length (source line 209, Stacks tag 0EA7); invoked at proof line 277. Supplies the additivity input used to cut the problem down to the two special cases; the explicit short exact sequence and the additivity of length(M) - length(N) are inside the counted span.

Source reference. Original lemma 0EA7: sources/stacks-chow/pinned-original-source.tex, source lines 209–209. Same-chapter pointer invoked at proof line 277 to split the claim into the two cases M = 0 and phi surjective; the additivity remark, the displayed short exact sequence and the reduction are retained verbatim in the counted span.

Lemma 0.1. *Let R be a ring. If (M, N, φ, ψ) is a 2-periodic complex such that M, N have finite length, then $e_R(M, N, \varphi, \psi) = \mathit{length}_R(M) - \mathit{length}_R(N)$. In particular, if (M, φ, ψ) is a $(2,1)$-periodic complex such that M has finite length, then $e_R(M, \varphi, \psi) = 0$.*

Proof. Observe that on the category of 2-periodic complexes with M, N of finite length the quantity "$\mathrm{length}_R(M) - \mathrm{length}_R(N)$" is additive in short exact sequences (precise statement left to the reader). Consider the short exact sequence

$$0 \to (M, \mathrm{Im}(\varphi), \varphi, 0) \to (M, N, \varphi, \psi) \to (0, N/\mathrm{Im}(\varphi), 0, 0) \to 0$$

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The initial remark combined with the additivity of Lemma 0EA7 reduces us to the cases (a) $M = 0$ and (b) φ is surjective. We leave those cases to the reader. $\square$